TRAINER KEY. DO NOT DISTRIBUTE. Day 3, Session 3: The analyst is in the room (analyst clinic)

Reading the draft note: N1 correctly reads the pilot point estimate (1,014) but does not settle the 1,000 AED rule (interval 862 to 1,167 spans it). N2 correctly says all four methods show reductions but overstates "independent confirmation": the three city analyses reuse the same records and compare different things (E2). N3 acknowledges limits, then treats the pilot point estimate as enough for national approval. Table T9: pilot 1,014 (862 to 1,167); city extra change 812 (792 to 833); cut-off 721 (498 to 943); matching 1,029 (859 to 1,200). Pilot and matching intervals span 1,000; DiD and cut-off intervals lie below it; all lie above zero. **Analyst evidence E1-E8** (the analyst's answers): E1 pilot headline (400 businesses, one district; no national representativeness evidence); E2 methods are not independent replications; E3 interval vs rule; E4 no earlier trends (one before observation); E5 pre-programme manager age jump of about 5.3 years at the cut-off; no score-manipulation audit; E6 matching: 585 distinct controls, one reused 440 times, residual gaps 3.0 years age and 161 AED cost; E7 synthetic complete records, no real fieldwork audit; E8 benefit-cost ratio 1.86 is a conditional model (812 AED provisional saving; 1,800 AED cost; five years assumed). **Rating:** accept bounded next-phase advice or requests for evidence if source, uncertainty and decision consequences are clear; do not score only the sign-off button. Rate answers against the actual question.

Underlined = claim the source supports. ~~Struck out~~ = claim the source does not support. Green boxes = model answers (accept equivalent wording).

Page 1 | H9-A: draft decision note

To: Director, programme investment
Subject: GreenWaste expansion decision
Status: Draft for review; fictional training case

Read the recommendation, its qualifications and Table T9 before recording your own advice. This note is separate from the report you will read on Day 4.

N1. Recommendation to the director: approve phased national expansion of GreenWaste. The programme subsidises business waste-management technology. In the pilot district, the lottery comparison estimated an annual saving of 1,014 AED per business after 12 months, exceeding our 1,000 AED minimum-saving rule. This is the clearest basis for proceeding.

N2. The city analyses also estimate lower costs for participants, using the difference between the two changes, the score cut-off and matched comparisons. Their figures differ, but the common direction across four methods provides independent confirmation that the programme works. Table T9 reports the estimates and their uncertainty.

N3. We recognise that the pilot is from one district and the cut-off result is local. The city model gives a benefit-cost ratio of 1.86 under stated cost and duration assumptions. National delivery should monitor costs and implementation, but the pilot point estimate clears our rule of at least a 1,000 AED reduction, so no further evidence is needed before approving the expansion.

Table T9 | Annual saving per business at 12 months

Source	Method	Estimated saving AED	95% interval for saving AED
E1 / E3	Pilot lottery	1,014	862 to 1,167
E2 / E4	City extra change	812	792 to 833
E2 / E5	Jump near score 58	721	498 to 943
E2 / E6	City matched comparison	1,029	859 to 1,200

Units: AED per business per year. Positive numbers mean reduced waste cost. Interval endpoints are sorted on that saving scale. **Decision rule:** minimum annual saving of **1,000 AED**.

The pilot is a separate district sample. The city difference between the two changes, the score cut-off and matching use the same city records; they make different comparisons. H9-B on page 2 describes them. Source IDs point to the analyst cards, supplied during the clinic.

Paragraph IDs N1-N3; table ID T9. Source: GreenWaste data. Each interval is the 95% interval for that method. Do not substitute this draft for a verified national appraisal.

Page 2 | H9-B: what was compared?

The four estimates concern annual waste cost at 12 months. A shared outcome and common sign do not make populations, comparisons or causal requirements interchangeable.

Source	Comparison	Causal requirement to scrutinise
E1	Pilot: 400 businesses in one district; lottery assigned 200 to support and 200 to comparison. Costs at 12 months.	Valid lottery and comparable outcome follow-up; no other systematic differences created by assignment.
E2 / E4	City: participant businesses' change (before to after) minus comparison businesses' change in the same 10,000-business file.	Without support, the groups would have had parallel cost trends; no differential concurrent changes.
E2 / E5	City: fitted cost jump at score 58, using 545 businesses within two score points on either side.	Without support, expected outcomes would vary smoothly at 58; no sorting or other discontinuous cause.
E2 / E6	City: 4,794 participant businesses paired to comparison businesses on four standardised pre-programme features; controls may be reused.	Conditional on the matched features, no remaining outcome-relevant selection; adequate comparable controls.

The city file has **10,000 businesses**, including **4,794 participants**. RDD uses a local subset; matching selects and may reuse city controls. City results are not independent replications of one national effect. The lottery pilot is a separate **400-business** sample.

Five families for useful questions

1. Who was compared, and how were they chosen?
2. What assumption supports a claim of cause?
3. Where do the data come from, and who is missing?
4. What are the estimate, interval and decision rule?
5. Does the recommendation go beyond the evidence?

Choose the questions that could change your decision. Specify the result or document you need, then say what you would do with the answer. A method name or a reassurance that the work is robust is not the requested evidence.

H9-B is a comparison map, not an assurance that these requirements hold. E1-E8 contain the supplied evidence and its limits. No fieldwork, prior-trend or score-manipulation records should be invented.

Page 3 | Opening decision and three questions

Q1 | My opening advice, before discussion

Sign off / With conditions / Request evidence
Reason and source: **N___ / T9 row ___**

> Accept "With conditions" or "Request evidence" with a reason that cites the source, e.g. "Request evidence: N1/T9 pilot row: the interval 862 to 1,167 spans the 1,000 AED rule and covers one district." "Sign off" is hard to justify: N3 says no further evidence is needed, but E1-E8 show gaps (see note).

Q2 | Our three specific evidence requests

Request 1: what should the analyst show? How could it change our advice?

> Example: request the national-representativeness and delivery-conditions evidence for the pilot (E1). If the pilot is not like the national population, the pilot estimate may not transfer, and I would not approve national expansion yet.

Follow-up if the first answer is incomplete:

> Example: "Which businesses or districts would expansion cover, and how do they differ from the 400 pilot businesses?"

Request 2: what should the analyst show? How could it change our advice?

> Example: ask for earlier cost trends for participants and comparison businesses (E4). If trends were similar before the programme, the DiD comparison becomes more plausible; the analyst should say the city file has only one before measurement.

Follow-up if the first answer is incomplete:

> Example: "If earlier trends are not available, what other evidence about concurrent changes affecting the two groups exists?"

Request 3: what should the analyst show? How could it change our advice?

> Example: ask for the full programme costs and persistence evidence (E8). If costs are higher or savings do not last five years, the 1.86 ratio and the case for national expansion could fall.

Follow-up if the first answer is incomplete:

> Example: "What is the full cost inventory, including staff, administration and maintenance, and what supports five years of savings?"

Our priority order: ___ / ___ / ___
Spokesperson: **(names; any roles)** **Listener/source checker:** **(names; any roles)**

Page 4 | Q3: record and assess actual answers

Log three substantial exchanges, including other groups' questions. Record what the analyst actually said, not what you hoped to hear.

Exchange 1

Question asked: **Example: "Is the 1,014 AED pilot result expected to apply nationally?" (to E1)**
Answer and source ID: E___ / N___ / T9

> Example: E1 answer: the note chose the pilot point estimate of 1,014 AED; 400 businesses, one district, lottery 200/200; no national representativeness evidence supplied. (Accept any log of what the analyst actually said with its ID.)

Supported / Partial / Unanswered: why? What follow-up or action?

> Rating example: Partial, because E1 explains where the number comes from but does not answer national reach. Follow-up: request evidence about the businesses and delivery conditions expected nationally (E1).

Exchange 2

Question asked: **Example: "Can you show earlier cost trends for the city difference-in-differences analysis?" (to E4)**
Answer and source ID: E___ / N___ / T9

> Example: E4 answer: the city file has one before observation and one at 12 months, so earlier trends cannot be checked; no earlier series supplied.

Supported / Partial / Unanswered: why? What follow-up or action?

> Rating example: Supported (honest disclosure): it answers the question by saying the check is not available. Action: request earlier cost measures or a documented review of concurrent changes.

Exchange 3

Question asked: **Example: "What checks have you done on the age jump at the 58 cut-off?" (to E5)**
Answer and source ID: E___ / N___ / T9

> Example: E5 answer: a five-point diagnostic shows pre-programme managers about 5.3 years younger below the cut-off; no score-manipulation audit supplied.

Supported / Partial / Unanswered: why? What follow-up or action?

> Rating example: Partial: it gives a concrete comparability concern but no audit resolving it. Follow-up: ask for the score-setting audit, balance checks and window sensitivity.

Rate the answer against its question. A stated limitation can be useful evidence without resolving the substantive request. "Not supplied" is preferable to an invented diagnostic.

Page 5 | Advice after the analyst clinic

Q4 | The unresolved issue that matters most

What evidence is needed? Cite the relevant source and explain its decision consequence.

> Example: the evidence for national reach (E1) and for full costs/persistence (E8): they decide whether pilot or modelled results apply to expansion. Accept any unresolved issue with a source ID and decision consequence.

If it supports the claim, I would:

> If supported: I would proceed to a limited next phase with a cost-monitoring and evaluation plan (not full national sign-off).

If it challenges the claim, I would:

> If it challenges: I would pause expansion, request the missing checks and revise the note.

Q5 | My final advice, before the AI exercise

Sign off / With conditions / Request evidence
Give one evidence-based reason and one boundary or condition.

> Example: "With conditions: approve a limited next phase because all four estimates show reductions (T9), on condition that expansion is evaluated and costs monitored, since the pilot and matching intervals span 1,000 and only one district is covered."

Compared with Q1, my reason changed / stayed the same because:

> Accept either, with a reason. An unchanged decision can reflect better reasoning; look for evidence use and limits, not agreement with the room.

Q6 | My independent request at the end

Close the AI output. Name the evidence to send, its purpose and how your advice could change. The original evidence may remain open; work alone.

> Example: "Send the pilot-versus-national population comparison (E1). Purpose: to judge whether the 1,014 estimate transfers. If not similar, I would limit the roll-out." Accept any specific request with a purpose and effect on advice. (Done alone with AI closed.)

An unchanged decision can reflect better reasoning. Your evidence use and limits matter, not agreement with a room vote. Complete Q6 after the Snapshot debrief.

Page 6 | Copilot Snapshot, Stage 1

8 minutes: mark up the supplied response, then compare with a partner. This response was written for the exercise. It is not a real AI answer.

The initial prompt

An evaluation of subsidised waste-management technology used a pilot lottery, city difference-in-differences, a score cut-off and matching. It reports annual savings in Table T9. List the critical questions a commissioner should ask before acting on these results.

The supplied response

> A commissioner should first check how the comparison groups were chosen, whether outcomes were measured consistently and whether the recommendation fits the population studied. Table T9 reports annual savings of 1,014, 812, 721, 1,029 AED. All intervals exclude zero; the pilot and matched intervals include the 1,000 AED minimum-saving rule. The point estimates alone do not settle that minimum.
>
> The pilot lottery offers a stronger basis for deciding whether GreenWaste caused the change than a simple before/after comparison. Earlier trends should be requested for the city difference-in-differences analysis, and comparability near the cut-off and after matching should be checked. The national recommendation also needs evidence on delivery costs and how long savings last.
>
> ~~However, four independent studies now agree on a reduction, which gives strong assurance that national expansion will achieve the intended saving. The large, complete city file establishes data reliability and leaves little reason to delay approval. Ask the evaluator to confirm that the results are robust, then proceed with national rollout.~~

Your source-based markup

Keep one useful question. Correct two unsupported assurances. Add one specific missing evidence request with a decision purpose. Cite H9-A/H9-B or E1-E8.

> Keep useful question: "how the comparison groups were chosen" (or the national-recommendation costs and savings persistence). Correct: "four independent studies agree ... strong assurance" (the three city analyses share records and compare different things, E2/H9-B); "the large, complete city file establishes data reliability" (E7: complete synthetic records do not show real fieldwork quality); "confirm that results are robust, then proceed" (generic reassurance is not evidence). Add: request earlier cost trends (E4) so that the DiD result can be judged before approval.

One source ID and the claim it checks:

> Example: E2 (city analyses reuse records; not independent replications) checks the claim "four independent studies".

Do not open the paper alternative on page 8 until Stage 1 is finished.

Page 7 | Copilot Snapshot, Stage 2

7 minutes: use the improved prompt in Copilot if available and approved. Answer its clarifying questions briefly, then verify the output. Use only this fictional case, not workplace or personal data.

A decision-focused prompt

I advise on national GreenWaste expansion with a 1,000 AED annual-saving rule. Use only H9-A (N1-N3, Table T9), H9-B and evidence E1-E8 supplied below. Ask me questions about the decision, population and acceptable uncertainty before giving advice. For each design, identify its main causal assumption, evidence that would challenge it and what is not supplied; do not claim one assumption is sufficient. Separate positive estimates, uncertainty about the minimum saving, national reach and modelled value for money. Recommend a provisional action and one evidence request that could change it. Cite source IDs. Do not invent diagnostics or treat shared city records as independent studies. [Paste the named sources.]

Source bundle: worksheet H9-A/H9-B, pages 1-2, plus analyst evidence bank E1-E8, pages 1-4, now supplied after the clinic. The draft note is a claim to review; the cards identify supplied evidence and unavailable checks.

If access or response time fails, use the prepared paper alternative on page 8. The task and seven-minute limit stay the same. No second chat is required.

Compare with the supplied response

I used: Copilot / prepared paper alternative
One useful revision, with its source:

> Accept any verified revision with a source, e.g. the Copilot answer separates the pilot from the city studies (E2) or admits earlier trends are not available (E4).

One claim or omission I still need to check, and how:

> Example: whether a cited source ID exists; whether it invents diagnostics; check against E1-E8 and H9-B.

Did the answer admit unavailable evidence and connect a request to advice?

> Yes if it says that evidence is unavailable (not invented) and connects a request to advice, e.g. "limited next phase with evaluation and cost monitoring; request target-population comparison (E1)".

AI can help formulate questions or conditional advice. Its output is not an additional study, diagnostic or source of evidence. Use the shared AI good-practice card if needed.

Page 8 | Prepared paper alternative

Hold until Stage 1 is finished. Use for Stage 2 when Copilot is unavailable, slow or not approved. This response was written for the exercise. It is not a real AI answer.

Possible clarifying questions

What population and delivery conditions does expansion cover? What risk of failing to reach the 1,000 AED saving rule is acceptable? Is the main concern whether GreenWaste caused the change, national reach, data quality or total spending?

A conditional response after the sources are supplied

> Before advising, I need to know the intended expansion population and acceptable risk of falling below 1,000 AED. T9 / E3 supports reductions on the supplied comparisons, but pilot and matching intervals span that rule. E2 says city methods share records and compare different populations. E4-E6 identify causal requirements and missing checks; earlier trends would inform, not prove, the DiD counterfactual. E7 does not establish real-world data quality. E8 gives a conditional model, not verified national value for money. One defensible provisional response is a limited next phase with an evaluation and cost-monitoring plan, rather than signing the note as written. Request a target-population and delivery comparison with the pilot (E1); it could change how far the pilot result can guide expansion. If whether GreenWaste caused the change, or the full costs, are the main concern instead, prioritise a different specific request and explain why.

Verify it, rather than accepting it

Locate each cited source. Decide whether the selected evidence request addresses your group's main concern. A different provisional action can be defensible if it states its limits.

A useful revision compared with Stage 1:

> Examples: the answer separates pilot from city analyses and cites sources; it recommends a provisional action; it asks about population and acceptable risk first.

An omission or remaining claim I would check:

> Examples: whether T9/E3 and E1 are cited correctly; whether E7 is read as not establishing real data quality; whether E8 is treated as a conditional model.

Record your comparison on page 7. Do not copy this response into the independent Q6 request.